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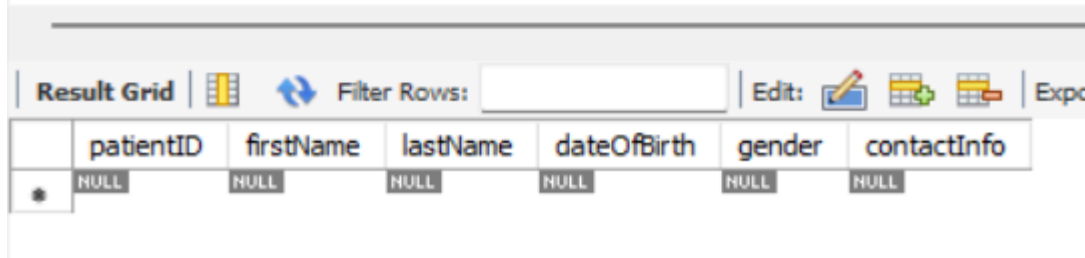
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University ST4005CEM Database System

Giriraj Rawat

1. Write a CREATE TABLE statement for the Patient table with all fields shown in the ERD. Include appropriate data types and a primary key.

```
3
4 • create table patient (
5     patientID int primary key,
6     firstName varchar(20),
7     lastName varchar(20),
8     dateOfBirth date,
9     gender enum("male","female","others"),
10    contactInfo char(10)
11 );
12 • select * from patient;
```



The screenshot shows a database management tool interface. At the top, there is a SQL editor with a CREATE TABLE statement for the 'patient' table. The statement defines columns: patientID (int, primary key), firstName (varchar(20)), lastName (varchar(20)), dateOfBirth (date), gender (enum with values 'male', 'female', 'others'), and contactInfo (char(10)). Below the editor, there is a 'Result Grid' tab. The grid shows the structure of the 'patient' table with columns: patientID, firstName, lastName, dateOfBirth, gender, and contactInfo. The first row of data is all NULL values.

	patientID	firstName	lastName	dateOfBirth	gender	contactInfo
*	NULL	NULL	NULL	NULL	NULL	NULL

2. Create the Appointment table with foreign keys referencing Patient and Doctor. Add a CHECK constraint so Status can only be 'Scheduled', 'Completed', or 'Cancelled'.

```
--
22 • create table appointment (
23     appointment int primary key,
24     patientID int,
25     doctorID int,
26     appointmentDate date default(curdate() ),
27     reason varchar(50),
28     status varchar(20) check (status in ("Scheduled","Completed","Cancelled"))
29 );
30
31 • alter table appointment add constraint patient_id_fk foreign key (patientID) references patient(patientID) ;
32 • desc appointment;
33
34 • alter table appointment add constraint doctor_id_fk foreign key (doctorID) references doctor(doctorID);
35
```

3. Create the Prescription table with all ERD fields. Ensure PrescriptionID auto-increments and both PatientID and DoctorID are foreign keys.

```
create table prescription (  
  prescriptionID int primary key auto_increment,  
  patientID int,  
  doctorID int,  
  medication varchar(20),  
  dosage varchar(20),  
  startDate date,  
  endDate date,  
  foreign key (patientID) references patient(patientID),  
  foreign key (doctorID) references doctor(doctorID)  
);
```

4. Insert a new doctor named Dr. raman khatiwada, specialty 'Cardiology', phone '9812311222', email 'khatiwada@hospital.com' into the Doctor table.

```
48 • insert into doctor (doctorID,firstName,lastName,speciality,phone,email)  
49   values( 1,"Dr. raman"," khatiwada","Cardiology","9812311222","khatiwada@hospital.com");  
50 • select * from doctor;  
51  
52
```

Result Grid					
Filter Rows:					
Edit:   					
Export/Import:  					
Wrap Cell Content: 					
doctorID	firstName	lastName	speciality	phone	email
1	Dr. raman	khatiwada	Cardiology	9812311222	khatiwada@hospital.com
NULL	NULL	NULL	NULL	NULL	NULL

5. Insert a patient named Ram Sharma, born 1990-05-10, gender 'M', into the Patient table. Then insert a billing record for him with Amount 5000 and Status 'Pending'.

```
65 • select * from patient;
```

```
66 • select * from billing;
```

Result Grid | Filter Rows: | Edit: | E

	patientID	firstName	lastName	dateOfBirth	gender	contactInfo
▶	101	Ram	Sharma	1990-05-10	male	9812345678
*	NULL	NULL	NULL	NULL	NULL	NULL

```
67 • select * from billing;
```

```
68 • truncate billing;
```

Result Grid | Filter Rows: | Edit: | E

	billingID	patientID	amount	billindDate	status
▶	1	101	5000.00	2026-05-08	Pending
*	NULL	NULL	NULL	NULL	NULL

6. Insert a medical record for PatientID 101 with diagnosis 'Hypertension', treatment 'Medication', recorded today.

```
80 • insert into medicalRecord (patientID, diagnosis,treatment, Notes)
```

```
81 values(101,"Hypertension","Medication","Follow the instruction");
```

```
82 • select * from medicalRecord;
```

```
83
```

Result Grid | Filter Rows: | Edit: | Export/Import: | V

	recordID	patientID	diagnosis	treatment	recordDate	Notes
▶	1	101	Hypertension	Medication	2026-05-08	Follow the instruction
*	NULL	NULL	NULL	NULL	NULL	NULL

7. Update the Status of all appointments for PatientID 101 from 'Scheduled' to 'Completed'.

```
85 • insert into appointment (appointmentID,patientID,doctorID,reason,status)
86 values(1,101,1,"All good","Scheduled");
87 • update appointment set status="COMPLETED" where patientID=101;
88
```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap C

	appointmentID	patientID	doctorID	appointmentDate	reason	status
▶	1	101	1	2026-05-08	All good	COMPLETED
*	NULL	NULL	NULL	NULL	NULL	NULL

8. A doctor changed their email. Update Doctor email to 'ramankhatiwada@hospital.com' where DoctorID = 1.

```
89 • update doctor set email='ramankhatiwada@hospital.com' where doctorID=1;
90
91
```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell

	doctorID	firstName	lastName	speciality	phone	email
▶	1	Dr. raman	khatiwada	Cardiology	9812311222	ramankhatiwada@hospital.com
✱	NULL	NULL	NULL	NULL	NULL	NULL

9. A prescription's EndDate needs to be extended by 30 days for PrescriptionID = 15. Write the UPDATE statement.

```
113
114 • update prescription set endDate = DATE_ADD(endDate, INTERVAL 30 DAY) where prescriptionID=15;
115
```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	prescriptionID	patientID	doctorID	medication	dosage	startDate	endDate
▶	13	101	1	Amoxicillin	500mg twice	2025-01-01	2025-02-10
	14	2	2	Ibuprofen	400mg thrice	2025-01-05	2025-02-14
	15	3	4	Metformin	850mg twice	2025-01-10	2025-03-21
	16	4	5	Atorvastatin	20mg once	2025-01-15	2025-03-16
	17	5	6	Cetirizine	10mg once	2025-01-20	2025-03-01
	18	6	3	Paracetamol	650mg thrice	2025-02-01	2025-03-12
✱	NULL	NULL	NULL	NULL	NULL	NULL	NULL

10. Delete all appointments that have Status 'Cancelled'.

115 • `select * from appointment;`

116 • `INSERT INTO appointment VALUES`

Result Grid

Filter Rows:

Edit:

Export/Import:

	appointmentID	patientID	doctorID	appointmentDate	reason	status
▶	1	101	1	2026-05-08	All good	COMPLETED
	2	101	2	2025-01-10	Chest pain checkup	Completed
	3	2	3	2025-01-15	Headache and dizziness	Completed
	4	3	4	2025-01-20	Knee joint pain	Completed
	5	4	5	2025-02-01	Child fever	Completed
	6	5	6	2025-02-10	Skin rash treatment	Scheduled
	7	101	3	2025-02-20	Follow up on headache	Cancelled
	8	2	4	2025-03-01	Back pain review	Cancelled
	9	3	5	2025-03-10	Vaccination checkup	Cancelled
	10	4	6	2025-03-15	Allergy consultation	Cancelled
	11	5	2	2025-03-20	Heart rate monitoring	Scheduled
✱	NULL	NULL	NULL	NULL	NULL	NULL

```

115 • select * from appointment;
116 • set sql_safe_updates=0;
117 • delete from appointment where status="cancelled";
118 • INSERT INTO appointment VALUES

```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wr

	appointmentID	patientID	doctorID	appointmentDate	reason	status
▶	1	101	1	2026-05-08	All good	COMPLETED
	2	101	2	2025-01-10	Chest pain checkup	Completed
	3	2	3	2025-01-15	Headache and dizziness	Completed
	4	3	4	2025-01-20	Knee joint pain	Completed
	5	4	5	2025-02-01	Child fever	Completed
	6	5	6	2025-02-10	Skin rash treatment	Scheduled
	11	5	2	2025-03-20	Heart rate monitoring	Scheduled
✱	NULL	NULL	NULL	NULL	NULL	NULL

11. Add a new column 'BloodGroup' of type VARCHAR(5) to the Patient table.

```
130 • alter table patient add column BloodGroup varchar(5);
131
```

Result Grid

Filter Rows:

Edit:

Export/Import:

	patientID	firstName	lastName	dateOfBirth	gender	contactInfo	BloodGroup
▶	2	Priya	Thapa	1985-07-22	female	9812345678	NULL
	3	Bikash	Rai	2000-11-08	male	9823456789	NULL
	4	Sita	Gurung	1995-01-30	female	9834567890	NULL
	5	Rohan	Magar	1978-06-12	others	9845678901	NULL
	6	Aarav	Sharma	1990-03-15	male	9801234567	NULL
	101	Ram	Sharma	1990-05-10	male	9812345678	NULL
✱	NULL	NULL	NULL	NULL	NULL	NULL	NULL

12. Rename the 'ContactInfo' column in the Patient table to 'Phone' and change its data type to VARCHAR(15).

```
127 • alter table patient rename column ContactInfo to Phone;
128 • select * from doctor;
129 • select * from patient;
130 • select * from prescription;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

	patientID	firstName	lastName	dateOfBirth	gender	Phone	BloodGroup
▶	2	Priya	Thapa	1985-07-22	female	9812345678	NULL
	3	Bikash	Rai	2000-11-08	male	9823456789	NULL
	4	Sita	Gurung	1995-01-30	female	9834567890	NULL
	5	Rohan	Magar	1978-06-12	others	9845678901	NULL
	6	Aarav	Sharma	1990-03-15	male	9801234567	NULL
	101	Ram	Sharma	1990-05-10	male	9812345678	NULL
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

```
128 • alter table patient modify Phone varchar(15);
129 • desc patient;
130 • select * from doctor;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	Field	Type	Null	Key	Default	Extra
▶	patientID	int	NO	PRI	NULL	
	firstName	varchar(20)	YES		NULL	
	lastName	varchar(20)	YES		NULL	
	dateOfBirth	date	YES		NULL	
	gender	enum('male','female','others')	YES		NULL	
	Phone	varchar(15)	YES		NULL	
	BloodGroup	varchar(5)	YES		NULL	

13. Add a NOT NULL constraint to the Billing.Amount column and set a default value of 0.

```
130 • desc billing;
131 • alter table billing modify amount decimal(10,2) not null default 0;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Field	Type	Null	Key	Default	Extra
▶	billingID	int	NO	PRI	NULL	auto_increment
	patientID	int	YES	MUL	NULL	
	amount	decimal(10,2)	NO		0.00	
	billindDate	date	YES		curdate()	DEFAULT_GENERATED
	status	enum('Pending','Completed')	YES		NULL	

```
131 • alter table billing modify amount decimal(10,2) not null default 0;
132 • SHOW CREATE TABLE billing;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Table	Create Table
▶	billing	CREATE TABLE `billing` (`billingID` int NOT NULL AUTO_INCREMENT, `patientID` int DEFAULT NULL, `amount` decimal(10,2) NOT NULL DEFAULT '0.00',

14. List each patient's full name along with their doctor's full name and appointment date.

```
144 • SELECT CONCAT(patient.firstName," ",patient.lastName) as patientName,
145          CONCAT(doctor.firstName," ",doctor.lastName) as doctorName ,
146          appointmentDate from appointment
147          join patient on patient.patientID=appointment.patientID
148          join doctor on doctor.doctorid=appointment.doctorID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	patientName	doctorName	appointmentDate
▶	Ram Sharma	Dr. raman khatiwada	2026-05-08
	Ram Sharma	Rajesh Karki	2025-01-10
	Rohan Magar	Rajesh Karki	2025-03-20
	Priya Thapa	Anita Shrestha	2025-01-15
	Bikash Rai	Suresh Pandey	2025-01-20
	Sita Gurung	Mina Tamang	2025-02-01
	Rohan Magar	Dipak Adhikari	2025-02-10

15. Show all patients and their billing amounts. Include patients with no billing records.

```

162
163 • select patient.* ,billing.amount from patient left join billing on patient.patientID= billing.patientID ;

```

patientID	firstName	lastName	dateOfBirth	gender	Phone	BloodGroup	amount
2	Priya	Thapa	1985-07-22	female	9812345678	NULL	1500.00
3	Bikash	Rai	2000-11-08	male	9823456789	NULL	2300.50
4	Sita	Gurung	1995-01-30	female	9834567890	NULL	NULL
5	Rohan	Magar	1978-06-12	others	9845678901	NULL	3200.00
5	Rohan	Magar	1978-06-12	others	9845678901	NULL	500.00
6	Aarav	Sharma	1990-03-15	male	9801234567	NULL	950.25
6	Aarav	Sharma	1990-03-15	male	9801234567	NULL	3800.00
101	Ram	Sharma	1990-05-10	male	9812345678	NULL	5000.00
101	Ram	Sharma	1990-05-10	male	9812345678	NULL	4100.00

16. Retrieve all prescriptions along with the prescribing doctor's name and the patient's name.

```

165 • select prescription.* ,CONCAT(patient.firstName," ",patient.lastName) as patientName ,
166      CONCAT( doctor.firstName," ",doctor.lastName) as doctorName from prescription
167      join patient on prescription.patientID= patient.patientID
168      join doctor on prescription.doctorid=doctor.doctorID;

```

prescriptionID	patientID	doctorID	medication	dosage	startDate	endDate	patientName	doctorName
13	101	1	Amoxicillin	500mg twice	2025-01-01	2025-02-10	Ram Sharma	Dr. raman khatiwada
14	2	2	Ibuprofen	400mg thrice	2025-01-05	2025-02-14	Priya Thapa	Rajesh Karki
15	3	4	Metformin	850mg twice	2025-01-10	2025-03-21	Bikash Rai	Suresh Pandey
16	4	5	Atorvastatin	20mg once	2025-01-15	2025-03-16	Sita Gurung	Mina Tamang
17	5	6	Cetirizine	10mg once	2025-01-20	2025-03-01	Rohan Magar	Dipak Adhikari
18	6	3	Paracetamol	650mg thrice	2025-02-01	2025-03-12	Aarav Sharma	Anita Shrestha

17. Find all patients born between 1980 and 2000 using the BETWEEN operator.

```
168 join doctor on prescription.doctorId=doctor.doctorId;
169 • select * from patient where dateOfBirth between "1980-01-01" and "2000-01-01";
```

	patientID	firstName	lastName	dateOfBirth	gender	Phone	BloodGroup
▶	2	Priya	Thapa	1985-07-22	female	9812345678	NULL
	4	Sita	Gurung	1995-01-30	female	9834567890	NULL
	6	Aarav	Sharma	1990-03-15	male	9801234567	NULL
	101	Ram	Sharma	1990-05-10	male	9812345678	NULL
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

18. List all doctors whose specialty is either 'Cardiology' or 'Neurology' using the IN operator.

```
169 • select * from patient where dateOfBirth between "1980-01-01" and "2000-01-01";
170 • select * from doctor where speciality in ('Cardiology','Neurology');
```

	doctorID	firstName	lastName	speciality	phone	email
▶	1	Dr. raman	khatiwada	Cardiology	9812311222	ramankhatiwada@hospital.com
	2	Rajesh	Karki	Cardiology	9851234567	rajesh@gmail.com
	3	Anita	Shrestha	Neurology	9862345678	anita@gmail.com
*	NULL	NULL	NULL	NULL	NULL	NULL